

A proof of Conjecture 00000000037

Sidon sets of primes of size $(N/\log N)^{1/2-o(1)}$

1 The conjecture and the reading

Conjecture 00000000037 reads (the Chinese text says the same):

Conjecture: Among the primes there exists a Sidon set of size $(N/\log N)^{1/2-o(1)}$.

Definition 1. A finite set $A \subseteq \mathbb{N}$ is a *Sidon set* if all pairwise sums $a + b$ ($a, b \in A$, $a \leq b$) are different; equivalently, $a + b = c + d$ with $a, b, c, d \in A$ forces $\{a, b\} = \{c, d\}$ as multisets (so $a = b$ is allowed). This is the definition on the Wikipedia page *Sidon sequence* (quoted in the attempt record). In Lean: `C37.IsSidon`.

Reading. The text names N without defining it; the factor $N/\log N \sim \pi(N)$ indicates that N bounds the primes. We read the conjecture as follows: *there are Sidon sets A_N , every element of which is a prime $\leq N$, and a function $\delta(N) \rightarrow 0$, with $|A_N| = (N/\log N)^{1/2-\delta(N)}$ for all large N .* Equivalently, for every real $\varepsilon > 0$ and every sufficiently large N ,

$$(N/\log N)^{1/2-\varepsilon} \leq |A_N| \leq (N/\log N)^{1/2+\varepsilon}. \quad (*)$$

We prove both forms. The upper half of $(*)$ holds for *every* Sidon set of naturals $\leq N$, so the exponent $1/2$ cannot be improved. The one-sided reading (“for every $\varepsilon > 0$ and all large N there is a Sidon set of primes $\leq N$ of size $\geq (N/\log N)^{1/2-\varepsilon}$ ”) follows from $(*)$. So does the reading “among the first N primes” (lower bound), since a prime $\leq N$ is one of the first N primes.

Conventions. $\log$ is the natural logarithm and x^y the real power; $\pi(n)$ is the number of primes $\leq n$; $\lfloor N/2 \rfloor$ is written $N/2$. Statements are “for all large N ”. For small N the sets A_N may be empty; this does not matter for an eventual statement.

Not covered. We do not treat a reading in which one *infinite* Sidon set A of primes is required, with counting function $|A \cap [1, N]| \geq (N/\log N)^{1/2-o(1)}$ for all N . The text says “of size”, which refers to a finite set depending on N . The infinite reading would give an infinite Sidon set of integers with $A(x) > x^{1/2-o(1)}$. Wikipedia (*Sidon sequence*) records that statement as a conjecture of Erdős.

2 Proof

Lemma 2 (Lean: `et_sidon`, `T_sidon`, `T_card`). *Let p be an odd prime and $e(k) = 2pk + (k^2 \bmod p)$ for $0 \leq k < p$. If $e(k_1) + e(k_2) = e(k_3) + e(k_4)$, then $\{k_1, k_2\} = \{k_3, k_4\}$. Hence for every shift b the set $T_{p,b} = \{e(k) + b : k < p, e(k) + b \text{ prime}\}$ is a Sidon set of primes, and $k \mapsto e(k) + b$ is injective. Moreover $e(k) < 2p^2$.*

Proof. Write $r_i = k_i^2 \bmod p \in [0, p)$. Then $e(k_1) + e(k_2) = (r_1 + r_2) + 2p(k_1 + k_2)$ with $0 \leq r_1 + r_2 < 2p$, and the same for k_3, k_4 . Dividing by $2p$ gives $k_1 + k_2 = k_3 + k_4$, hence $r_1 + r_2 = r_3 + r_4$. In $\mathbb{Z}/p$ this gives $k_1 + k_2 = k_3 + k_4$ and $k_1^2 + k_2^2 = k_3^2 + k_4^2$. Substituting $k_2 = k_3 + k_4 - k_1$ gives $2(k_1 - k_3)(k_1 - k_4) = 0$. Since p is odd, $k_1 \equiv k_3$ or $k_1 \equiv k_4 \pmod{p}$. All $k_i < p$, so $k_1 = k_3$ or $k_1 = k_4$, and the other equality follows. A sum identity for $T_{p,b}$ reduces to one for e after cancelling $2b$. Finally $e(k) \leq 2p(p-1) + p - 1 < 2p^2$. $\square$

Lemma 3 (Lean: `exists_shift`). *Let $p \geq 1$ and $L \geq 0$. Some $b \leq L$ satisfies $p(\pi(L) - 2p^2) \leq (L+1) \cdot \#\{k < p : e(k) + b \text{ prime}\}$ (truncated subtraction).*

Proof. Put $K = 2p^2$. For each $k < p$ we have $e(k) < K$, and $q \mapsto q - e(k)$ maps the primes $q \in [K, L]$ injectively to shifts $b \leq L$ with $e(k) + b$ prime. Their number is at least $\pi(L) - K$. Summing over k and exchanging the sums gives $\sum_{b \leq L} \#\{k < p : e(k) + b \text{ prime}\} \geq p(\pi(L) - K)$. There are $L+1$ shifts, so one of them gets at least the average. $\square$

Lemma 4 (Lean: `construct`). *If $x := \pi(N/2) \geq 64$, there is a Sidon set A of primes $\leq N$ with $x^3 < 64N^2|A|^2$.*

Proof. Let $L = N/2$ and $m = \lfloor \sqrt{x/16} \rfloor \geq 2$. Bertrand's postulate (Mathlib `Nat.exists_prime_lt_and_le_two_mul`) gives a prime p with $m < p \leq 2m$. So $p \geq 3$ is odd, $4p^2 \leq 16m^2 \leq x$, and $16p^2 \geq 16(m+1)^2 > x$. Take b from Lemma 3 and $A = T_{p,b}$. Its elements are primes $< 2p^2 + L \leq (L+1)/2 + L \leq N+1$ (using $x \leq L+1$). With $c = |A|$, Lemma 3 and $x - 2p^2 \geq x/2$ give $px \leq 2(L+1)c \leq 2Nc$. Hence $x^3 < 16p^2x^2 = 16(px)^2 \leq 64N^2c^2$. $\square$

Lemma 5 (Lean: `pi_lower`). *For $0 < \eta \leq 1$ and all large N : $4N^{1-\eta} \leq \pi(N/2)$.*

Proof. Mathlib's `Chebyshev.pi_ge` states $\pi(n) \geq (n \log 2 - \log(n+1))/\log n$. Since $\log x = o(x^\eta)$ and $\log x = o(x)$ (Mathlib `isLittle0_log_rpow_atTop`, `isLittle0_log_id_atTop`), for large real x we have $12x^{1-\eta} \log x + \log(x+1) \leq x \log 2$. With $n = N/2$ this gives $\pi(n) \geq 12n^{1-\eta}$. Also $N \leq 3n$, so $N^{1-\eta} \leq 3^{1-\eta}n^{1-\eta} \leq 3n^{1-\eta}$. $\square$

Lemma 6 (Lean: `sidon_card_sq`, `sidon_upper`). *If $A \subseteq \{0, \dots, N\}$ is Sidon, then $|A|^2 - |A| \leq 2N$. Hence for every $\varepsilon > 0$ and all large N , every Sidon set of naturals $\leq N$ has $|A| \leq (N/\log N)^{1/2+\varepsilon}$.*

Proof. The map $(a, b) \mapsto a - b$ on ordered pairs with $a \neq b$ is injective into $[-N, N] \setminus \{0\}$: if $a - b = c - d$ then $a + d = c + b$, so $(a, d) = (c, b)$ or $a = b$, and $a = b$ is excluded. This gives $|A|^2 - |A| \leq 2N$. With $|A| \leq N+1$ we get $|A|^2 \leq 4N$ for $N \geq 3$. Since $(\log x)^{1+2\varepsilon} = o(x^{2\varepsilon})$ (Mathlib `isLittle0_log_rpow_rpow_atTop`), eventually $4N \leq N^{1+2\varepsilon}/(\log N)^{1+2\varepsilon} = (N/\log N)^{1+2\varepsilon}$. $\square$

Theorem 7 (Lean: `C37.conjecture_37`). *There are Sidon sets A_N of primes $\leq N$ such that $(*)$ holds for every real $\varepsilon > 0$ and all large N .*

Proof. Let A_N be the set of Lemma 4 when $\pi(N/2) \geq 64$, and $\emptyset$ otherwise. Since $\pi(n) \rightarrow \infty$, the first case holds for all large N . The upper half of $(*)$ is Lemma 6. For the lower half, replace ε by $\varepsilon' = \min(\varepsilon, 1/2)$; this is allowed because $N/\log N \geq 1$. Apply Lemma 5 with $\eta = 2\varepsilon'/3$, so $x = \pi(N/2) \geq 4N^{1-\eta}$. Then $64N^{3-3\eta} \leq x^3 < 64N^2|A_N|^2$, so $|A_N|^2 > N^{1-2\varepsilon'}$ and $|A_N| > N^{1/2-\varepsilon'} \geq (N/\log N)^{1/2-\varepsilon'}$ (since $\log N \geq 1$ for $N \geq 3$ and $1/2 - \varepsilon' \geq 0$). $\square$

Corollary 8 (Lean: `C37.conjecture_37_little0`). *For the same sets, $\delta(N) = \frac{1}{2} - \log|A_N|/\log(N/\log N)$ tends to 0, and $|A_N| = (N/\log N)^{1/2-\delta(N)}$ for all large N .*

Proof. For large N we have $N/\log N > 1$ and $|A_N| > 0$. Taking logarithms in (*) with $\varepsilon/2$ gives $|\delta(N)| \leq \varepsilon/2$. The identity is $B^{\log c / \log B} = c$ for $B > 1, c > 0$. $\square$

Corollary 9 (Lean: `C37.conjecture_37_firstPrimes`). *Each A_N consists of primes p_k (the k -th prime, `Mathlib Nat.nth Nat.Prime k`) with $k < N$, and $|A_N| \geq (N/\log N)^{1/2-\varepsilon}$ for all large N .*

Proof. A prime $a \leq N$ equals p_k with $k = \#\{\text{primes} < a\} \leq a - 1 < N$ (`Mathlib Nat.nth_count`). $\square$

3 Lean formalization

File `lean/Conjecture37/Basic.lean` (418 lines, only `import Mathlib`, Lean v4.33.1 / `Mathlib v4.33.1`). Main statement:

```
theorem conjecture_37 :
  exists A : Nat -> Finset Nat,
    (forall N, (forall a in A N, a.Prime /\ a <= N) /\ IsSidon (A N)) /\
    forall eps : Real, 0 < eps -> forall^f N : Nat in atTop,
      ((N : Real) / log N) ^ (1/2 - eps) <= (A N).card /\
      ((A N).card : Real) <= ((N : Real) / log N) ^ (1/2 + eps)
```

(ASCII transcription of the Lean source). Besides it, `conjecture_37_little0` gives the literal $\delta(N) \rightarrow 0$ form, `conjecture_37_firstPrimes` the first- N -primes form, and `sidon_upper` the universal upper bound. `lake build` succeeds, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for all four. There is no `sorry`, `native_decide` or added axiom.